COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/29/26	**DATE SUBMITTED:**	7/29/26
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ACTIVITY NO. 4 **ARRAY BASED VS STL STACK IMPLIMANTATION**

1. PROCEDURE OUTPUT

Objective

- Implement a Stack Data Structure using two distinct approaches in C++: a custom Static Array implementation and the C++ Standard Template Library

Tasks & Intended Learning Outcomes (ILOs):

- **ILO A:** Implement a stack using the C++ STL (`std::stack`).
- **ILO B.1:** Implement a stack using a Static Array.

Submission Requirements:

- **Documentation:** Please use the official **Activity Template** for your write-up.
- **Source Code:** Submit your C++ file named as `LastName_Seatwork4_1.cpp`.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

LastName_Seatwork4_1.cpp | main.cpp

HOA4p1 > ILOA > LastName_Seatwork4_1.cpp > main()

```cpp
#include <iostream>
#include <stack>

int main() {

    // push
    // pop
    // isEmpty
    // isFull
    // Top
    // create
    std::stack<char> stack1;

    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;
    stack1.push('C');
    std::cout << "current top of the stack: " << stack1.top() << std::endl;
    stack1.push('Y');
    std::cout << "current top of the stack: " << stack1.top() << std::endl;
    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;
    std::cout << "What is the size of the stack? " << stack1.size() << std:

    // swap
    std::stack<char> stack2;
    stack2.push('C');
    std::cout << "\n--- Swapping stack1 and stack2 ---" << std::endl;
    stack1.swap(stack2);
    std::cout << "stack1 top after swap: " << stack1.top() << std::endl;
    std::cout << "stack2 top after swap: " << stack2.top() << std::endl;

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

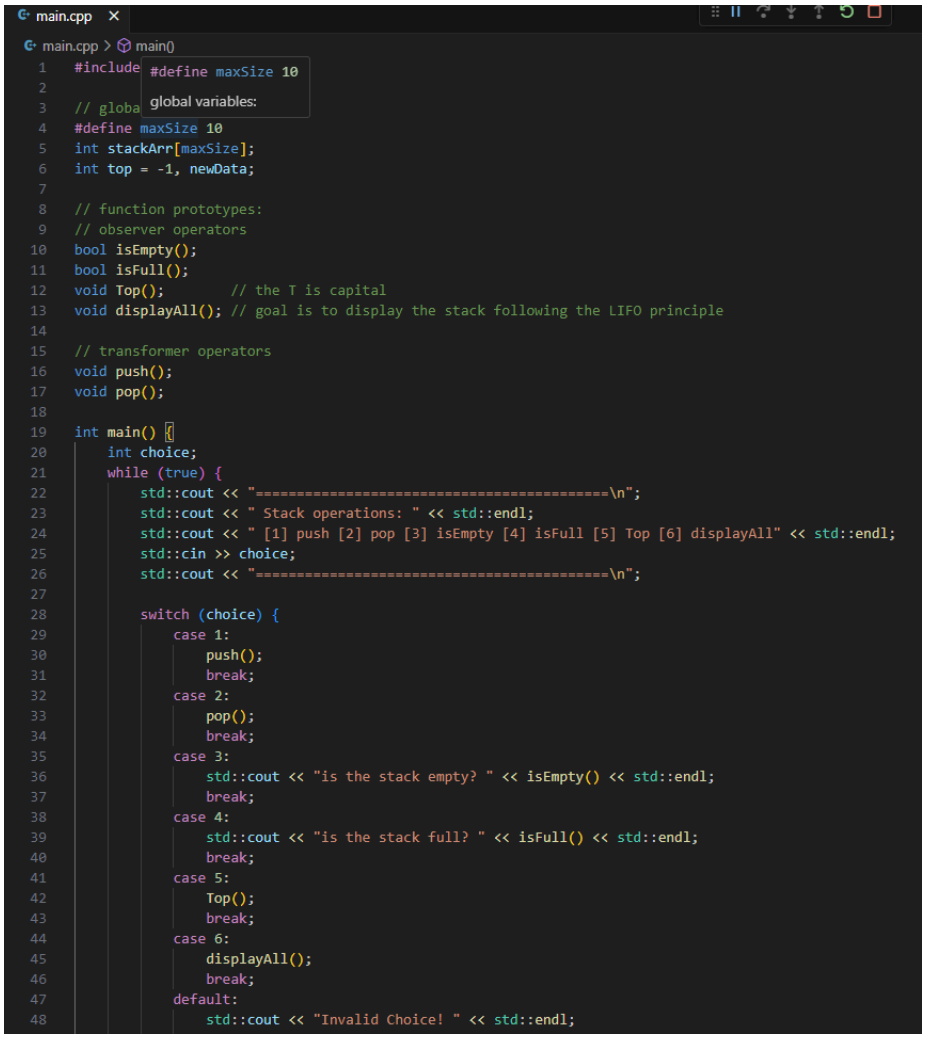
```
ne-Pid-piyh5lmo.xos' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Is the stack empty? 1
current top of the stack: C
current top of the stack: Y
current top of the stack: Y
Is the stack empty? 0
What is the size of the stack? 2

--- Swapping stack1 and stack2 ---
stack1 top after swap: C
stack2 top after swap: Y

--- Emplacing element ---
current top of stack1 after emplace: J
size of stack1 after emplace: 2
size of stack1 after emplace: 2
PS C:\Users\TIPQC\OneDrive\Documents\CPE12S4 - Reyes>
```

```
29
30      // emplace
31      std::cout << "\n--- Emplacing element ---" << std::endl;
32      stack1.emplace('J');
33      std::cout << "current top of stack1 after emplace: " << stack1.top() << std::endl;
34      std::cout << "size of stack1 after emplace: " << stack1.size() << std::endl;
35
36      return 0;
37  }
```

ILOB1



```
                break;
            }
        }

        return 0;
    }

    // function definitions
    bool isEmpty() {
        return (top == -1);
    }

    bool isFull() {
        return (top == maxSize - 1);
    }

    void Top() {
        if (isEmpty()) return;
        std::cout << "The top element is: " << stackArr[top] << std::endl;
    }

    // goal is to display the stack following the LIFO principle
    void displayAll() {
        if (isEmpty()) {
            std::cout << "Stack is empty!" << std::endl;
            return;
        }
        std::cout << "Stack elements (LIFO order): ";
        for (int i = top; i >= 0; i--) {
            std::cout << stackArr[i] << " ";
        }
        std::cout << std::endl;
    }

    void push() {
        // error catching:
        if (isFull()) {
            std::cout << "Stack Overflow!" << std::endl;
            return;
        }

        // ask the user what they want to insert:
        std::cout << " What number do you want to push? ";
        std::cin >> newData;

        // storing to the stack:
        stackArr[++top] = newData;
    }

    void pop() {
        // error catching:
        if (isEmpty()) return;

        // display the item to be popped:
        std::cout << "Popping: " << stackArr[top] << std::endl;

        // we remove the element from the stack
        top--;
    }
```

```
PS C:\Users\TIPQC\OneDrive\Documents\CPE12S4 - Reyes\ILOB1>  & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cp
' '--stdout=Microsoft-MIEngine-Out-3auou0cl.amc' '--stderr=Microsoft-MIEngine-Error-xc502is2.oto' '--pid=Micro
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
1
========================================
 What number do you want to push? 20
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
1
========================================
 What number do you want to push? 30
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
2
========================================
Popping: 30
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
1
========================================
 What number do you want to push? 35
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
1
========================================
 What number do you want to push?
5
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
5
========================================
The top element is: 5
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
6
========================================
Stack elements (LIFO order): 5 35 20
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
5
========================================
The top element is: 5
========================================
 Stack operations:
 [1] push [2] pop [3] isEmpty [4] isFull [5] Top [6] displayAll
```

The program successfully implements an array-based stack adhering strictly to the Last In, First Out (LIFO) principle. It incorporates all fundamental operations—Push, Pop, Empty, Full, and Display—to manage and visualize data. Execution outputs confirm that each operation functions as intended, with the top element and current stack size dynamically updating following every state change.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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